

Tableau Basics → Advanced Tableau & Review

This material focus on **business analytics and visual storytelling** - not just software buttons. For every visualization, ask:

- What question am I answering?
- Which chart answers it?
- What business decision can this insight support?

Learning Objectives

By the end of this module, you should be able to:

1. Access and use Tableau Public for hands-on practice.
2. Explain dimensions, measures, rows, columns, and the Marks card.
3. Connect Excel data and build basic visualizations in Tableau.
4. Choose appropriate charts for comparison, trend, relationship, and geographic analysis.
5. Create calculated fields, filters, and parameters.
6. Build interactive dashboards with filters and actions.
7. Apply dashboard design principles for management reporting.
8. Compare Power BI and Tableau using the same dataset and analytical workflow.

Topics Covered

Module	Focus
Getting Started	Tableau Public access and interface
Data Concepts	Dimensions, measures, connecting Excel
Basic Charts	Bar, line, KPI, profit analysis, scatter plot
Advanced Charts	Dual axis, heat map, treemap, map, Top-N, parameters
Dashboards	Layout, filters, actions, design principles
Integration	Power BI + Tableau comparison
Capstone	Retail performance analysis and presentation

1. How to Access Tableau

Tableau Public (Recommended for Practice)

Tableau Public is a free platform for creating and sharing visualizations online. Browser-based authoring is available - no software download required.

Important Limitation

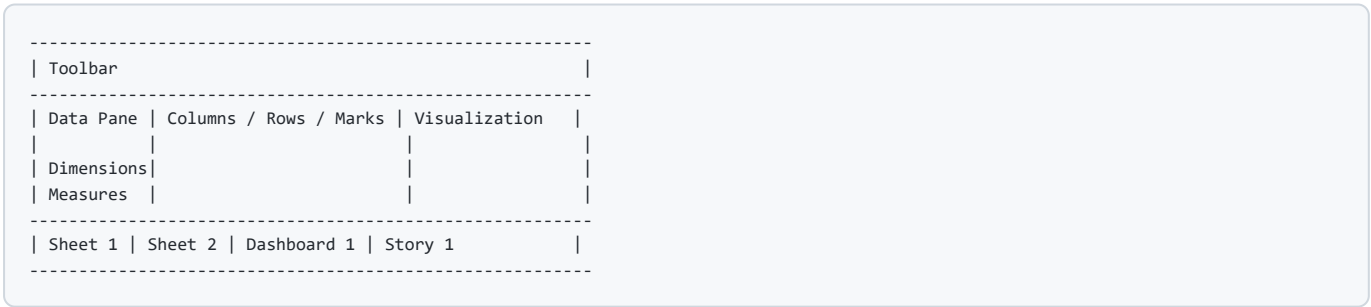
Tableau Public is for **public data only**. Published work can be publicly accessible. Do **not** upload confidential company, customer, employee, or financial data.

Setup Steps

- 1. Open [Tableau Public](#)
- 2. Select **Sign Up / Create Free Profile**
- 3. Create a free account (personal email is fine)
- 4. Sign in
- 5. Select **Create**
- 6. Choose **Web Authoring**

Note: Tableau Cloud and Tableau Server are organizational products with different sign-in requirements.

2. Tableau Interface



Area	Purpose
Data Pane	Dimensions, measures, calculated fields, parameters
Columns	Horizontal structure of the visualization
Rows	Vertical structure of the visualization
Marks Card	Color, size, label, detail, tooltip, shape
Sheets / Dashboards / Stories	Individual views and combined layouts

3. Core Analytical Framework

Question → Data → Visualization → Insight → Business Decision

Step	Example
Business question	Which product category generates the most revenue?
Data	Product Category + Sales
Visualization	Bar chart
Insight	<i>(Your observation from the data)</i>
Business decision	<i>(What management could investigate or act on)</i>

4. Dimensions and Measures

Dimensions

Categorical or descriptive fields.

Examples: Region, Product, Customer, Department, Gender, Category, City

Measures

Numerical fields that can be aggregated and analyzed.

Examples: Sales, Profit, Quantity, Revenue, Salary, Discount, Cost

Quick Check

Field	Dimension or Measure?
Region	
Sales	
Category	
Profit	
Customer Segment	
Quantity	

5. Connecting Excel Data

Practice Dataset - Sales.xlsx

Order Date	Region	Category	Product	Sales	Profit	Quantity	Customer Segment
01-Jan-26	South	Technology	Laptop	85000	12000	2	Corporate
05-Jan-26	North	Furniture	Chair	18000	4000	5	Consumer
10-Jan-26	West	Technology	Monitor	32000	6500	3	Corporate
15-Jan-26	East	Office Supplies	Paper	9000	1800	10	Consumer

Steps to Connect

1. Open Tableau Public
2. Create a new workbook
3. Select **Microsoft Excel** and choose `Sales.xlsx`
4. Review the data source: column names, data types, missing values, dates, numeric fields
5. Open **Sheet 1**

Workflow: Connect → Review/Prepare → Build views

6. Chart Selection Guide

Business Need	Recommended Chart
Compare categories	Bar chart
Show trend over time	Line chart
Show part-to-whole	Pie / Treemap
Show a single KPI	Card / KPI
Investigate relationship between numbers	Scatter plot
Compare two metrics over time	Dual axis
Compare region × category performance	Heat map
Show geographic performance	Map
Show top contributors	Top-N bar chart

7. Basic Visualizations

7.1 Sales by Region (Bar Chart)

Question: Which region generates the highest sales?

Field	Placement
Region	Columns
Sales	Rows
Sales	Label
Region	Color

Your analysis:

- Highest sales region:
 - Lowest sales region:
 - Possible reasons for differences:
 - What should management investigate?
-

7.2 Sales Trend (Line Chart)

Question: How has sales changed over time?

Field	Placement
Order Date	Columns
Sales	Rows
Category	Color

Adjust date level: Year, Quarter, Month, Week, Day

Your analysis:

- Is sales growth consistent?
 - Which category shows seasonal behavior?
 - Any unusual peaks or drops?
-

7.3 Sales by Category (Bar Chart)

Field	Placement
Category	Rows
Sales	Columns

Sort descending.

Question: Which category contributes most to revenue?

7.4 Profit Analysis

Field	Placement
Category	Rows
Profit	Columns
Profit	Color

Key concept:

High Sales $\neq$ High Profit

Your analysis:

Category	Sales Level	Profit Level	Notes

7.5 Sales vs Profit (Scatter Plot)

Question: Is higher sales associated with higher profit?

Field	Placement
Sales	Columns
Profit	Rows
Category	Color
Product	Detail

Your analysis:

- Products with high sales but low profit:
- Products with low sales but high margins:
- Products needing further investigation:

8. Calculated Fields

Calculated fields let you build business metrics from existing data.

Example: Profit Margin

1. Analysis → Create Calculated Field

2. Name: Profit Margin

3. Formula:

$\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}])$

4. Format as percentage

Question: Which category has the highest profit margin?

Category	Profit Margin

9. Advanced Visualizations

9.1 Dual Axis - Sales and Profit Over Time

Question: Is increasing revenue also producing increasing profit?

Step	Action
1	Order Date → Columns; Sales → Rows; Profit → Rows
2	Right-click second axis → Dual Axis
3	Synchronize axes if needed
4	Sales → Bar; Profit → Line

Your observation:

9.2 Heat Map - Region × Category

Question: Which regions and categories perform strongly?

Field	Placement
Region	Rows
Category	Columns
Sales	Color
Profit	Label

Marks type: **Square**

Complete your interpretation:

Region	Technology	Furniture	Office Supplies
South			
North			
West			
East			

9.3 Treemap - Contribution to Sales

Question: What categories/products contribute most to sales?

Field	Placement
Category	Label
Sales	Size
Sales	Color

Larger rectangle = greater contribution

9.4 Geographic Map

Field	Placement
State	Map
Sales	Size
Profit	Color

Question: Which geographical markets generate high sales?

9.5 Top-N Analysis

Question: What are the top 10 products by revenue?

1. Product → Rows; Sales → Columns
2. Product filter → **Top** → **By field** → Top 10 by Sum of **Sales**

Reflection: Should the company focus only on these products? Consider revenue concentration, profitability, and long-tail products.

9.6 Parameters - Switch Metrics

Create a parameter to switch between Sales, Profit, and Quantity.

Parameter: Metric Selection

Calculated field:

```
CASE [Metric Selection]
WHEN "Sales" THEN SUM([Sales])
WHEN "Profit" THEN SUM([Profit])
WHEN "Quantity" THEN SUM([Quantity])
END
```

Use the calculated field in your chart instead of building three separate visuals.

9.7 Filters

Add interactive filters for Region, Category, Customer Segment, and Order Date.

1. Drag field to **Filters**
 2. Select **Show Filter**
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10. Dashboard Creation

A **dashboard** combines multiple worksheets into one analytical view.

Worksheets to Build First

Sheet	Content
1	Total Sales (KPI)
2	Total Profit (KPI)
3	Sales by Region
4	Sales Trend
5	Sales by Category
6	Sales vs Profit

Suggested Layout

SALES KPI PROFIT KPI PROFIT MARGIN KPI
SALES TREND
SALES BY REGION SALES BY CATEGORY
SALES vs PROFIT

Dashboard Filters

Add and test: Region, Category, Customer Segment, Date

- Select a region → charts should update
- Select a category → dashboard should respond

Dashboard Actions

Action Type	What It Does
Filter	Clicking one chart filters others
Highlight	Clicking highlights a category across charts
URL	Opens an external webpage from a selection

11. Dashboard Design Principles

Avoid

- Too many charts and colors
- Huge legends and excessive text
- No clear KPI or business purpose

Recommended Flow

KPI → Trend → Comparison → Drivers → Action

Four Questions Every Dashboard Should Help Answer

1. What happened?
 2. Where did it happen?
 3. Why might it have happened?
 4. What should management investigate?
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12. Executive Dashboard Reference

Example structure for a **Sales Performance Dashboard**:

Component	Examples
KPIs	Total Sales, Total Profit, Profit Margin, Orders
Charts	Monthly trend, Sales by Region, Sales by Category, Profit vs Sales, Top 10 Products
Filters	Year, Region, Category, Customer Segment

Sample Dashboard Link: [Sample Sales Dashboard](#)

13. Power BI + Tableau Practice

Use the **same dataset** in both tools.

Build in Both Tools

Component	Power BI	Tableau
KPI - Total Sales	☐	☐
Sales by Region	☐	☐
Monthly Sales Trend	☐	☐
Profit by Category	☐	☐
Sales vs Profit	☐	☐
Filters (Region, Category, Date)	☐	☐

Comparison Worksheet

Feature	Power BI	Tableau
Data connection		
Bar chart		
Line chart		
Filters		
Dashboard		
Calculated metric		
Interactivity		
Geographic analysis		
Ease of exploration		
Business storytelling		

Reflection Questions

1. What is similar between both tools?
 2. What is different?
 3. Which concepts transfer directly?
 4. Why is data preparation important in both?
 5. Why can the same data tell different business stories?
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14. Key Terms

Term	Definition
Workbook	Container for your Tableau analysis
Worksheet	A single visualization
Dashboard	Multiple worksheets combined for analysis
Story	A sequence of visualizations for narrative
Dimension	Categorical field (e.g., Region, Product)
Measure	Numerical field (e.g., Sales, Profit)
Calculated field	A field created using a formula
Parameter	An interactive control that changes calculations or views
Filter	Limits data shown in a view or dashboard
Dual axis	Two measures on one chart with separate axes
Top-N filter	Shows only the top (or bottom) ranked values

15. Revision Summary

Fundamentals

Workbook, Worksheet, Dashboard, Story, Dimensions, Measures, Rows, Columns, Marks, Filters

Visualizations

Bar chart, Line chart, Scatter plot, Heat map, Treemap, Map, Dual-axis chart, KPI

Analytics Skills

Calculated fields, Aggregation, Top-N, Parameters, Filters, Sorting, Trend analysis

Business Analytics Applications

Sales analysis, Profitability, Customer segmentation, Regional analysis, Product analysis, Correlation, KPI monitoring, Data storytelling

Key Takeaway

“What business question am I trying to answer?”

Choose the data fields and visualization that best answer that question. The same analytical thinking applies in both Tableau and Power BI.

Flow: data → visualization → insight → business decision

Reference Links

- [Tableau Public](#)
- [Tableau Public Support & Getting Started](#)